Curriculum Vitae

Miles V. Vollmer

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PROFESSIONAL SUMMARY

AI and performance engineer with 3+ years of professional experience. I build, integrate, and optimize enterprise systems, and my results are on the record. At Moviri I lead cloud migration, BI platform modernization, and automation work for global banking and telecommunications clients. My work has cut $2M+ per month in infrastructure spend, won an Innovation Award for an enterprise dashboard, and re-platformed 450 dashboards (7,000+ visualizations) single-handedly with AI-assisted tooling I architected. Trained in Northeastern's CS/AI program, I work in any language or stack, from front end to back end, and my machine learning work spans peer-reviewed research, reinforcement learning, and production multi-agent systems. I find and deliver improvements at every scale, from a single query to an enterprise platform.

EDUCATION & CERTIFICATIONS

B.S. in Computer Science with AI Concentration, Northeastern University, Boston, MA (2024)

Certifications: AWS Cloud Practitioner; BMC Certified Associate (2025); BMC AIOps (2026: Helix Operations Management, Intelligent Integrations, Log Analytics, Service Monitoring, Dashboards)

TECHNICAL SKILLS

Engineering: Full-stack development, system integration, ETL pipeline design, RESTful API integration and reverse-engineering, performance and cost optimization, cloud capacity planning, observability, CI/CD and release automation, test-driven development

Languages: Python, Java, SQL (Redshift, PostgreSQL, Databricks SQL), TypeScript, JavaScript, R, Bash, HTML/CSS. Language-agnostic by training; productive in any stack.

Cloud & Data Platforms: Databricks (AI/BI Dashboards, Unity Catalog, SQL Warehouses, Lakehouse Federation), AWS (EC2, S3, Redshift), Google Cloud Run, Grafana, Tableau, Power BI, dbt, Streamlit, Docker, Kubernetes, Git, GitHub Actions, Dynatrace, BMC Helix (Continuous Optimization, Operations Management), NetSuite

AI & Machine Learning: LLM agents and multi-agent systems, RAG, Google ADK, Gemini, AI-assisted development (Claude Code, context engineering), PyTorch, TensorFlow, HuggingFace, scikit-learn, NLP, reinforcement learning (CFR/MCCFR, ReBeL, Q-learning, MCTS), ensemble machine learning and SHAP pipelines (published)

Science & Bioinformatics: Snakemake, ANGSD, population genomics (FST, SMC++, LFMM), 16S rDNA analysis, GWAS, SLURM/HPC

PROFESSIONAL EXPERIENCE

Performance Engineering Consultant – Moviri *Jan 2025 – Present*

Consulting engineer to global banking and telecommunications clients. Engagements below span cost optimization, full-stack development, platform migration, and AI-assisted automation, from analysis through production.

Senior Consultant, Major UK Bank: Enterprise Capacity Platform Migration and Optimization

- Led capacity analysis across ~10,000 overprovisioned servers; rightsizing cut infrastructure spend by **$2M+ per month**.
- Built the front and back end of a Business Units drill-down dashboard that navigates from the platform-wide view to the root cause for any business unit in a few clicks; winner of the **Innovation Award at the BMC Helix London Roadshow**.
- Responsible, with the solution architect, for nine of the migration's eleven workstreams.
- Redesigned the client's custom ETL framework for the new SaaS platform (the migration's largest workstream): added event and application data support, migrated all ServiceNow integrations, and promoted the ETLs to production.
- Designed a three-tier availability-zone resolution chain for a Dynatrace collector covering 2,300+ GCP hosts: DNS hostname parsing, then one cached bulk lookup per run, with per-host API calls only as a last resort.
- Built a Datamart Dependencies Extractor from scratch: three Java ETL modules mapping dashboard, datamart (memoization, cycle detection), and ETL dependencies; packaged as a documented company repository for reuse across clients.
- Adapted a cloud cost extractor to price current and recommended VM instances against the platform's catalogue, feeding cost-saving estimates into its dashboards.
- Built a GCP fleet dashboard suite with multi-level drill-down, rightsizing recommendation views across the full organizational hierarchy, and license chargeback reporting by business unit.
- Co-authored ~50 of the 74 content pages of the contractual Construction Document, including its largest chapter with a newly onboarded team member.
- Presented delivered work to client stakeholders; co-presented the follow-on Phase 1B proposal (public cloud coverage, ServiceNow risk events, efficiency recommendations).

Major Italian Telecommunications Provider: AWS to Databricks Analytics Migration

- Ran the planning phase independently for a ~100 TB migration: mapped ~300 in-scope objects across dbt, Grafana, and Streamlit, identified 71 independent migration clusters, and recommended a lowest-risk pilot batch.
- Produced effort estimates (8.7 person-days for the pilot, 32.5 for full scope) reconciled with the commercial proposal, and an AWS cost model for the cross-region data move ($89K/yr).
- Root-caused a federation permissions blocker affecting 87% of required data sources (41 of 47) with an escalation-ready diagnosis.
- Verified data coherence between the new Kafka-fed catalog and the legacy Redshift warehouse across five pipeline families; isolated the one gap to an upstream feed outside the migration.
- Sole engineer on the dashboard migration workstream: re-platforming **450 Grafana dashboards (7,000+ visualizations)** into Databricks on a fixed timeline. The schedule requires ~20 dashboards and ~335 visualizations per working day (one dashboard every ~20 minutes), before verification, rework, and republishing.
- Architected a 15-tool Python toolchain for audit, SQL transpilation, spec-driven assembly, numeric validation against source data, and dry-run-first delivery.
- Sustained this pace with AI-assisted engineering: deliberate context curation, a 176-entry decision log, and regression gates; **100+ dashboards in the first two weeks**.
- Absorbed scope growth beyond the original agreement, including 150+ source dashboards modified by the client after initial export, while working around the functional differences between the two platforms.
- Improved on the source dashboards' functionality; reverse-engineered undocumented Databricks AI/BI internals (conditional threshold coloring, dynamic titles, typography, legends) into reusable build capabilities protected by a 63-test regression suite, so each fix propagates to every remaining dashboard.
- Main point of contact in client review meetings for the workstream (~10 to date), turning feedback into changes delivered within days; authored handover, runbook, and design-rationale documentation for client stakeholders.

Internal AI and Automation

- Designed and piloted HOAO (Human-Oriented Automation Orchestration), a hierarchical multi-agent framework (Google ADK, Gemini, FastAPI) in which agents draft, sandbox-test, and validate their own tools under human approval, then cache them for reuse; telemetry-based routing sends each task to a cheaper or more capable model. Deployed on Google Cloud Run under Moviri's AI Labs program.
- Built a full-stack meal voucher application on NetSuite data, packaged for Windows and macOS through GitHub Actions, replacing hours of monthly manual HR work.
- Built an LLM onboarding agent (Google ADK) that emails new hires, parses replies, and requests missing information before HR verification.
- Delivered five internal knowledge-sharing presentations on AI automation and client project work.

Product and Platform Engineering

- Led a major rework of the Entuity REST integration connector (ORM refactor, regex filtering); top contributor to the repository by commits.
- Automated monthly product release packaging across all offerings with YAML CI/CD workflows.

Research Assistant – Evolutionary Genomics Lab, Northeastern University *May 2021 – Sep 2021, Sep 2022 – Aug 2024*

- Built ensemble machine learning and SHAP pipelines (Random Forest, SVM, Multilayer Perceptron, Lasso Regression, and others) predicting coral disease states from 16S rDNA sequencing data; **97%+ classification accuracy**.
- Identified the bacterial taxa most predictive of disease through SHAP feature-importance analysis, surfacing candidate pathogens for further study.
- Co-authored a peer-reviewed publication in *Environmental Microbiology* describing the pipeline and findings.

Data Research Analyst – Iafrate Lab, Massachusetts General Hospital (MGH) *Jan 2022 – Aug 2022*

- Built an automated Genome-Wide Association (GWA) pipeline for COVID-19 genetic data, improving large-scale processing efficiency.
- Visualized demographic variation in positivity rates; findings informed clinical intervention and policy discussions.

PUBLICATIONS

Selwyn, JD, Despard, BA, Vollmer, MV, Trytten EC, Vollmer, SV (2024). "Machine Learning based identification of putative coral pathogens in endangered Caribbean staghorn coral." *Environmental Microbiology* 26 (9): e16700. Link

Vollmer, M., Senthil Kumar, D. (2024). "A Multi-Method Approach to Training a Competitive No-Limit Texas Hold'em Poker Agent Using MCTS, CFR, and ReBeL Architectures." Northeastern University, CS 4180 Reinforcement Learning. Link

AWARDS & RECOGNITION

- Innovation Award, BMC Helix London Roadshow (2026), for the Business Units drill-down dashboard.

SELECTED PROJECTS

Coral Population Genomics Pipeline: Two-pass Snakemake/ANGSD pipeline across 290 whole genomes (2 species, 3 Caribbean regions), portable across workstation, SLURM HPC, and AWS. Retained 2,034,805 SNPs; FST analysis quantified near-complete species isolation, SMC++ reconstructed both species' demographic histories, and LFMM identified a candidate thermal-tolerance locus. Fixed a recombination-mapping bug blocking downstream analysis and cut compute costs ~70% with fault-tolerant spot instances, automated retry, and auto-stop. Feeds an in-preparation publication.

No-Limit Texas Hold'em Poker Agents: Implemented and benchmarked CFR, MCCFR, Q-learning, and a ReBeL-style self-play agent with neural policy, value, and bet-sizing networks for 6-player poker; the ReBeL agent consistently outperformed Q-learning agents across 10 trials of 1,000 rounds. Written up as a paper (above).

4-Player Chess AI: Minimax engine with move ordering, alpha-beta pruning, and search caching for a 2v2 chess variant on a 14x14 board; 100% win rate vs. random AI and about 50% vs. experienced human players.

Extracting Traffic Data with NLP: Pipeline over the Boston 311 API to extract and classify traffic violation reports for urban planning use.

Image Processing System: Java MVC application with a GUI supporting image filters, transformations, and format conversions.

OTHER INTERESTS

Soccer (Co-Captain of Northeastern FC in the Boston Metro Sunday Soccer League), surfing, scuba diving, chess, traveling, hiking